
Algorithm 1: Baseline label-setting algorithm for maximum country coverage

Input:

- V . Set of nodes
- E_s . Set of scheduled edges (v, u, dep, arr)
- E_t . Set of transfer edges (v, w, Δ)
- $country(v)$. Country associated with node v

Output:

- P . Sequence of labels representing an optimal route within 24 hours

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1: Initialize  $Labels(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize queue  $Q \leftarrow \emptyset$ 
3: Initialize  $best\_label \leftarrow null$ 
4: // Initialize labels at departures
5: for all  $v \in V$  do
6:   for all scheduled departure from  $v$  at time  $dep$  with arrival  $arr$  do
7:      $L \leftarrow (v, dep, \{country(v)\}, pred = null, start\_time = dep)$ 
8:      $Labels(v) \leftarrow Labels(v) \cup \{L\}$ 
9:      $Q \leftarrow Q \cup \{L\}$ 
10: // Main loop
11: while  $Q \neq \emptyset$  do
12:   extract  $L = (v, t, C, pred, start\_time)$  from  $Q$ 
13:   // Only consider labels within 24-hour journey span
14:   if  $t - L.start\_time > 24 : 00$  then
15:     continue
16:   if  $best\_label = null \vee |C| > |best\_label.C|$  then
17:      $best\_label \leftarrow L$ 
18:   // Extend via scheduled edges
19:   for all  $(v, u, dep, arr) \in E_s$  do
20:     if  $dep \geq t \wedge dep - L.start\_time \leq 24 : 00$  then
21:        $t_{new} \leftarrow arr$ 
22:       if  $t_{new} - L.start\_time > 24 : 00$  then
23:         continue
24:        $C' \leftarrow C \cup \{country(u)\}$ 
25:        $L' \leftarrow (u, t_{new}, C', pred = L, start\_time = L.start\_time)$ 
26:       if  $\nexists L'' \in Labels(u)$  such that
27:          $(L''.t \leq t_{new} \wedge L''.C \supseteq C' \wedge (L''.t < t_{new} \vee L''.C \neq C'))$  then
28:           remove all  $L'' \in Labels(u)$  such that
29:              $(t_{new} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{new} < L''.t \vee C' \neq L''.C))$ 
30:            $Labels(u) \leftarrow Labels(u) \cup \{L'\}$ 
31:            $Q \leftarrow Q \cup \{L'\}$ 
32:   // Extend via transfers
33:   for all  $(v, w, \Delta) \in E_t$  do
34:      $t_{new} \leftarrow t + \Delta$ 
35:     if  $t_{new} - L.start\_time > 24 : 00$  then
36:       continue
37:      $L' \leftarrow (w, t_{new}, C, pred = L, start\_time = L.start\_time)$ 

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36:   if  $\nexists L'' \in \text{Labels}(w)$  such that
       $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
37:     remove all  $L'' \in \text{Labels}(w)$  such that
       $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
38:      $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 
39:    $Q \leftarrow Q \cup \{L'\}$ 

40: // Reconstruct path
41:  $P \leftarrow$  empty sequence
42:  $L \leftarrow \text{best\_label}$ 
43: while  $L \neq \text{null}$  do
44:   prepend  $L$  to  $P$ 
45:    $L \leftarrow L.\text{pred}$ 
46: return  $P$ 

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